

AI Receptivity or AI Adoption Breadth?

A Tool-Specific Reanalysis of the Lower-Literacy/Higher-Usage Link

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(Replication and robustness reanalysis)

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Abstract

Recent evidence reported by [Tully et al. \(2025\)](#) suggests that lower artificial intelligence (AI) literacy predicts greater receptivity toward AI. We revisit this claim using the public data from Study 3 of that article, which measures past usage of five AI tool categories on a five-point frequency scale. We first reproduce the negative association between AI literacy and aggregate AI usage using OLS on participant-level averages, binary logit, ordered logit, and multinomial logit specifications. We then show that the aggregate relationship masks substantial heterogeneity by tool type. After covariate adjustment, AI literacy does not significantly predict text-AI usage (ordered-logit $\beta = -0.097$, $p = .357$), whereas it remains a strong predictor of non-text AI adoption ($\beta = -0.376$, $p < .001$). Binary, ordered-logit, and multinomial specifications further suggest that the non-text effect is primarily an adoption/non-adoption pattern rather than evidence of intensive use: the per-standard-deviation odds ratio of ever having used a non-text AI tool is 0.68. These results do not overturn the original finding, but they qualify its interpretation: lower AI literacy may predict broader adoption of certain AI-labelled tools rather than uniformly greater AI receptivity or text-AI use.

Keywords: AI literacy, AI adoption, ordinal regression, replication, robustness, construct validity, marketing analytics.

1 Introduction

The proliferation of generative AI products has renewed marketing scholars' interest in the individual-level antecedents of AI adoption ([Puntoni et al., 2021](#); [Hermann and Puntoni, 2024](#)). Among recent contributions, [Tully et al. \(2025\)](#) document a counterintuitive lower-literacy, higher-receptivity link across seven studies: consumers who score lower on objective tests of AI knowledge are, on average, more receptive to AI. Study 3 of that article is empirically pivotal because, unlike Studies 4 and 6, it uses reported past usage of real AI products, rather than stated preferences for hypothetical tasks, as the dependent variable. This makes it a more conservative test of the proposed relationship and, accordingly, a key target for replication and robustness checks.

The original Study 3 (N=401, Amazon Mechanical Turk) measures usage frequency over the previous six months for five AI tool categories: digital image generators (e.g., DALL-E), AI-powered productivity tools (e.g., Zapier), AI-powered design services (e.g., Canva), AI-powered

health apps (e.g., Headspace), and AI-powered writing assistants (e.g., ChatGPT). Each tool is rated on a five-point frequency scale ranging from Never to Weekly. The authors average these five items into an AI receptivity index and regress the index on AI literacy.

We see two reasons to revisit this specification. First, averaging ordinal five-point items into a single continuous index is a standard but contested practice, and the field has accumulated substantial methodological evidence that doing so can distort estimates of treatment effects and even invert their sign (Liddell and Kruschke, 2018; Agresti, 2010). Second, and more substantively, the five tool categories arguably tap heterogeneous behavioral constructs. Text AI (writing assistants) had reached mass diffusion by mid-2023 after the release of ChatGPT, whereas image generators, productivity bots, website builders, and meditation apps remained niche products. Aggregating them into a single “AI receptivity” index conflates intensive text-AI use with broader adoption of AI-labelled consumer tools.

In this paper we revisit Study 3 along both dimensions. We do not dispute the aggregate lower-literacy/higher-usage association; rather, we show that its substantive meaning changes when AI usage is decomposed by tool type and modelled as ordinal adoption data. The reanalysis makes three points:

1. Replication: the pooled Study 3 association is real. We reproduce the original direction under OLS, binary logit, ordered logit, and multinomial logit.
2. Measurement: averaging ordinal usage responses across heterogeneous AI categories compresses different behavioral processes into a single index.
3. Interpretation: the adjusted relationship is weak and non-significant for text AI alone but strong for non-text AI tools, and the non-text effect is concentrated at the adoption margin. Study 3 is therefore better interpreted as evidence about non-text AI adoption breadth rather than as evidence about general AI usage intensity or text-AI receptivity.

2 Background

Algorithm aversion and algorithm appreciation. A long literature in marketing and judgment-and-decision-making documents systematic individual responses to algorithmic agents. Dietvorst et al. (2015) show that people abandon algorithms after observing errors, even when the algorithm outperforms a human alternative. Logg et al. (2019) show the complementary phenomenon, algorithm appreciation, in numerical estimation tasks. Castelo et al. (2019) and Longoni and Cian (2022) qualify both patterns by showing that task type (objective vs. subjective; utilitarian vs. hedonic) moderates consumer responses, while Longoni et al. (2019) document resistance to medical AI driven by uniqueness neglect. Yalcin et al. (2022) show that the direction of the algorithmic decision (favorable vs. unfavorable) matters for consumer reactions, and de Bellis et al. (2023) show that the perceived meaning of manual labor impedes the adoption of autonomous products. Across this literature, contextual and task-related factors play a central role, and individual-level determinants of receptivity remain comparatively understudied.

AI literacy. The construct of AI literacy itself was articulated by Long and Magerko (2020) as a set of competencies needed to evaluate, communicate with, and use AI in everyday contexts. Subsequent reviews (Ng et al., 2021) adapted classical literacy frameworks to characterize AI-specific knowledge, use, evaluation, and ethics. Against this backdrop, Tully et al. (2025) make the novel claim that higher AI literacy decreases receptivity to AI, because demystifying AI dispels perceptions of magic and the awe such perceptions evoke.

Tool-level heterogeneity in AI adoption. Recent macro evidence underscores why aggregating AI tool categories is risky. McElheran et al. (2024) document that, as of 2018, fewer than 6% of U.S. firms used any of five core AI technologies, and adoption was strongly concentrated by industry and firm size. Brynjolfsson et al. (2025) show that productivity gains from generative AI assistants are heterogeneous across workers, with the largest gains accruing to lower-skill agents. Acemoglu and Restrepo (2020) make a parallel point for industrial robots. These results suggest that “AI adoption” is not a single behavioral phenomenon but a family of tool-specific decisions whose drivers may differ across tool categories.

Ordinal versus metric analysis. On the methodological side, McCullagh (1980) introduced the proportional-odds model that we adopt here, and Agresti (2010) provides a comprehensive treatment of ordinal regression. Liddell and Kruschke (2018) review hundreds of psychology articles and document that treating ordinal Likert responses as metric can yield false positives, false negatives, and, in extreme cases, sign inversions of estimated effects. Their recommendation, which we follow, is to estimate ordered-probit or ordered-logit models on the item-level data rather than OLS on averaged indices.

3 Original Study 3 and Analytic Concern

Tully et al. (2025) report that “Participants with lower AI literacy were more receptive to have AI complete the tasks” ($B = -0.09$, $SE = .02$, $t(399) = -5.73$, $p < .001$), with the coefficient remaining significant after controlling for technology readiness, general knowledge, motivation for autonomy, and gender ($B = -0.11$, $p < .001$). The dependent variable is the mean of five items $Y_{ij} \in \{1, 2, 3, 4, 5\}$ measuring usage frequency of image generators, productivity tools, website builders, health apps, and writing assistants.

Two features of the design motivate our reanalysis. First, the five items correspond to qualitatively different AI products. Figure 1 shows their empirical distributions: text AI is used at least occasionally by roughly two-thirds of respondents, while the four non-text categories are right-censored at the lowest category (“Never”), with 65–78% of respondents reporting no usage in the previous six months. Second, the outcome

$$Y_{ij} \in \{1, 2, 3, 4, 5\}$$

is ordinal, sparse for several tools, and repeated within participant. Averaging is not inherently invalid, but it conflates two distinct margins: (i) *adoption* (whether respondents use a tool at

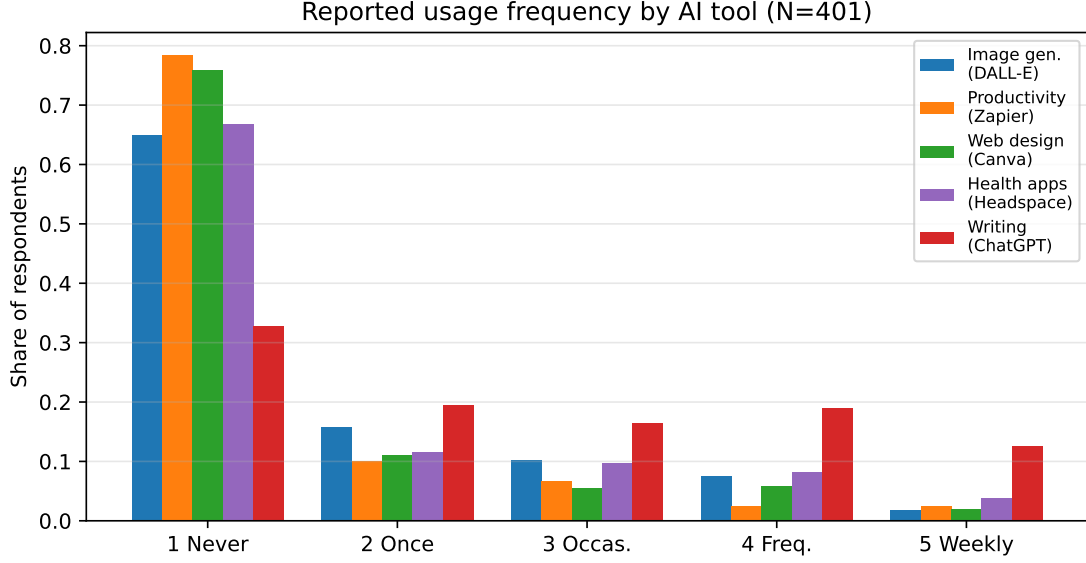


Figure 1: Reported usage frequency by AI tool category (Study 3, N=401). Text AI is the only category for which a clear majority of respondents are users; the four non-text categories are right-skewed with most respondents at category 1 (*Never used in the last six months*).

all) and (ii) *intensity* (how often they use it conditional on adoption). When those margins differ across tools, the pooled effect estimate need not have a clean substantive interpretation.

4 Reanalysis Strategy

We download the public Study 3 data from [ResearchBox #1491](#) and use the AI literacy score (SC0, summed correct answers on the 17-item AI-constructed measure), the five usage items, and the same covariates reported in the original Table 4 (age, income, general knowledge, motivation for autonomy, gender). We standardize the AI literacy score (and continuous covariates) so that coefficients are interpretable as the effect of a one-standard-deviation increase. We then fit four model families across three outcome groupings.

Model families.

1. OLS on the participant-level average. This is the closest match to the original Study 3 specification and serves as our pooled benchmark.
2. Binary logit on item-level data with task fixed effects. We consider two thresholds: $Y > 3$ (“frequent use”) and $Y > 1$ (“adoption”).
3. Ordered logit (proportional-odds model) ([McCullagh, 1980](#)). This is the natural model for an ordered Likert response with $K = 5$ categories and is robust to violation of metric-scale assumptions ([Liddell and Kruschke, 2018](#)).
4. Multinomial logit, treating the five categories as nominal. This is a stress test: it does not impose proportional odds or any ordering and is therefore robust to violations of either.

Table 1: Pooled five-tool replication of Study 3. Coefficient on a +1 standard-deviation increase in AI literacy, controlling for age, income, general knowledge, motivation for autonomy, and gender. Item-level models include task fixed effects. N participants = 401; N item-level observations = 2,005.

Model	Coef.	SE	<i>p</i> -value	Notes
OLS on participant average	−0.179	0.053	.001	HC3 SE
Binary logit, item-level ($Y > 3$)	−0.328	0.086	< .001	OR = 0.72
Ordered logit, item-level	−0.306	0.054	< .001	Prop. odds
Multinomial logit, item-level	−0.189 to −0.452	NA	< .05 all	vs. $y = 1$

Outcome groupings.

1. Pooled (5 tools): the original Study 3 outcome.
2. Text only: `AI_text` alone.
3. Non-text only: `AI_image`, `AI_productivity`, `AI_website`, `AI_healthapp`.

Implementation. All models are fit in Python with `statsmodels`. OLS and binary-logit standard errors are heteroskedasticity-robust (HC3). Item-level models include task fixed effects. Full code and a Jupyter notebook reproducing every number and figure in the paper are provided as supplementary material.

5 Results

5.1 Replication of the pooled effect

Table 1 reports the AI-literacy coefficient under each of the four model families on the pooled five-tool outcome. The estimated effect of a one-standard-deviation increase in AI literacy is negative and statistically significant in every specification. The OLS estimate ($\hat{\beta} = -0.179$, $p = .001$) recovers, after standardization, the result reported in the original paper. The ordered logit ($\hat{\beta} = -0.306$, $p < .001$) and binary logit at the scale midpoint ($\hat{\beta} = -0.328$, $p < .001$, odds ratio = 0.72) tell the same qualitative story. The multinomial logit (omitted from the table for compactness) yields the same direction across all four non-reference categories. We conclude, as the original authors did, that the pooled association is robust to model choice.

5.2 Text vs. non-text decomposition

Table 2 re-estimates the ordered-logit and binary-logit specifications separately on the text-AI outcome and on the four non-text outcomes. The contrast is clear. For text AI alone, the ordered-logit coefficient is small and not significantly different from zero ($\hat{\beta} = -0.097$, $SE = 0.105$, $p = .357$); the binary “frequent use” logit and the binary adoption logit are also non-significant. For non-text AI, the same coefficient is roughly four times larger in magnitude and highly significant ($\hat{\beta} = -0.376$, $SE = 0.063$, $p < .001$), and the result is consistent across the ordered, midpoint-binary, and adoption-binary specifications.

Table 2: Text vs. non-text decomposition. Coefficient on a +1 standard-deviation increase in AI literacy, with the same covariates as Table 1. Item-level models include task fixed effects.

Outcome / Model	Coef.	SE	<i>p</i>	Notes
Text AI only (N=401)				
OLS	−0.075	0.081	.353	HC3 SE
Binary logit ($Y > 3$)	−0.112	0.142	.433	OR = 0.89
Binary adoption ($Y > 1$)	−0.060	0.131	.648	OR = 0.94
Ordered logit	−0.097	0.105	.357	Prop. odds
Non-text AI (4 tools; 1,604 item-level obs.)				
OLS on participant avg.	−0.204	0.053	< .001	HC3 SE
Binary logit ($Y > 3$)	−0.404	0.104	< .001	OR = 0.67
Binary adoption ($Y > 1$)	−0.385	0.068	< .001	OR = 0.68
Ordered logit	−0.376	0.063	< .001	Prop. odds

The contrast is even sharper at the adoption margin. The binary adoption specification ($Y > 1$, i.e., “ever used in the last six months”) gives an odds ratio of 0.68 per one-standard-deviation increase in AI literacy for non-text tools, a clean, easily interpretable result, and an odds ratio almost indistinguishable from 1.0 for text AI.

Figure 2 visualizes the consequence of this asymmetry. We plot the ordered-logit-predicted probability $\Pr(Y = 1)$ (“Never used”) over the empirical range of standardized literacy, separately for the text and non-text models. For text AI, the predicted probability of non-use rises only modestly, from 0.27 at $z = -2$ to 0.36 at $z = +2$. For non-text AI, it rises sharply, from 0.50 to 0.82 across the same range. In other words, the higher a respondent’s AI literacy, the more likely they are to have never tried image generators, productivity bots, website builders, or health apps, while their probability of having used a ChatGPT-style writing assistant is essentially flat.

5.3 Coefficient summary

Figure 3 summarizes the AI-literacy coefficient across all three tool groupings and the three primary model families. Three patterns emerge.

First, all three model families agree within each tool grouping: the ordered logit, the midpoint binary logit, and the adoption binary logit give similar point estimates, and their confidence intervals overlap substantially. Methodological choice between metric and ordinal specifications is therefore not what drives our results.

Second, the magnitude of the AI-literacy effect for non-text tools ($\hat{\beta} \approx -0.38$) is roughly four times the magnitude for text AI ($\hat{\beta} \approx -0.10$), and the latter’s confidence interval crosses zero. The pooled coefficient (≈ -0.31) is an arithmetic compromise that obscures this asymmetry.

Third, for non-text tools the adoption coefficient ($Y > 1$, $\hat{\beta} = -0.385$) is essentially identical to the midpoint coefficient ($Y > 3$, $\hat{\beta} = -0.404$) and to the ordered-logit coefficient ($\hat{\beta} = -0.376$). This is what we would expect if the non-text effect operates almost entirely at the adoption margin: shifting respondents into and out of the lowest category “Never” is enough to explain the ordered and midpoint estimates.

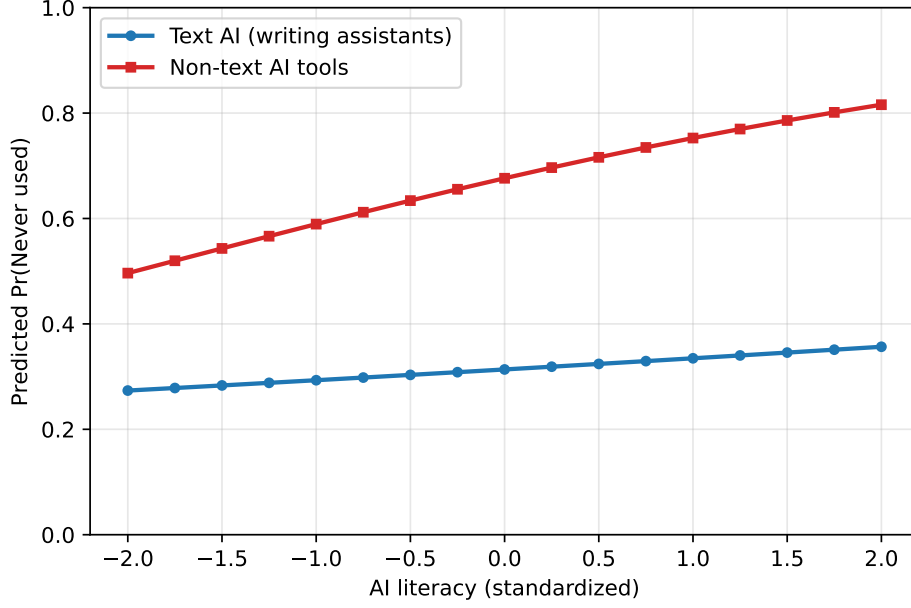


Figure 2: Predicted probability of non-use by AI literacy. Curves are derived from ordered-logit fits on text-AI only (blue) and on the four non-text AI tools (red, with task fixed effects), holding all covariates at their sample means. Higher AI literacy predicts much higher odds of having never used non-text AI tools; the relationship is roughly flat for text AI.

6 Discussion

The original aggregate association reported by [Tully et al. \(2025\)](#) is robust: it survives every reasonable parametric alternative to OLS on an averaged Likert index. What changes when the outcome is decomposed is its substantive interpretation. The pooled estimate combines three analytically distinct constructs:

1. General AI receptivity: a latent willingness to adopt AI across product categories.
2. Text-AI usage intensity: how often a respondent uses text-based generative AI tools such as ChatGPT, conditional on having adopted them.
3. Non-text AI adoption breadth: the probability of having tried at least one of several heterogeneous AI-labelled consumer products.

Our results suggest that Study 3 speaks most clearly to the third construct. The text-AI margin contributes essentially nothing to the aggregate effect: the slope of literacy on text-AI use is small, non-significant, and survives no model specification. The aggregate estimate is therefore mostly carrying information about adoption breadth across categories where the modal respondent has never used the product.

This re-interpretation does not contradict the broader theoretical claim that lower-literacy consumers may perceive AI as magical ([Tully et al., 2025](#)). But it does change what that claim implies for managers. The original article’s managerial recommendation is to “target consumers with lower AI literacy” as a growth segment. Our results narrow this prescription in two ways. First, the targeting case is much stronger for low-penetration, non-text AI product

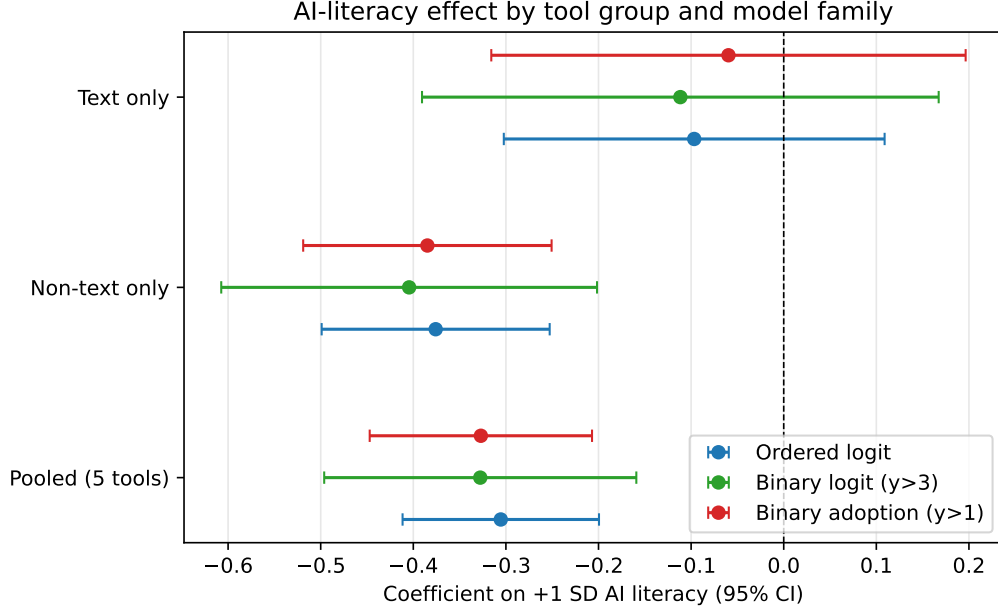


Figure 3: AI-literacy effect by tool group and model family. Markers are point estimates of the coefficient on a +1 standard-deviation increase in AI literacy; horizontal bars are 95% confidence intervals. For text AI alone, the coefficient is statistically indistinguishable from zero under every model family. For non-text AI, the coefficient is consistently negative and significant.

categories (image generation, productivity automation, AI design services, AI wellness apps) than for already-mass-market text assistants. Second, the non-text effect is driven primarily by the adoption margin rather than by intensive use: low-literacy consumers are differentially more likely to try these products, not necessarily to use them more intensively conditional on trial. This matters for product strategy. Lifetime-value calculations that assume a uniform usage-intensity response to literacy may overestimate the value of low-literacy acquisitions.

Methodologically, our reanalysis illustrates the value of treating heterogeneous ordinal items as what they are. Averaging five items with sharply different empirical distributions into a single “receptivity” index is convenient, and in this case it does not invert any sign, but it does flatten a multi-tool adoption pattern into something that reads like a single dispositional trait. The construct-validity question that follows, whether “AI receptivity” is one thing or several, deserves attention beyond the present application (Long and Magerko, 2020; Ng et al., 2021).

7 Limitations

Several limitations of our reanalysis warrant explicit mention.

- No new data. We rely entirely on the data and codebook released by the original authors. Their Studies 1, 2, and 4–7 use different measures of receptivity (cross-country adoption readiness, propensity to use generative AI for assignments, and relative preference for AI vs. human task completion) for which our critique does not directly apply.
- Observational and self-reported. Like the original Study 3, our analyses are correlational and rely on self-reported usage. Any measurement error in self-reported usage that

correlates with literacy could bias the estimated coefficients.

- Post hoc analysis. Our decomposition is not preregistered. We have nevertheless implemented the four model families exactly as described in our analytic strategy and provide all code.
- No causal claim. We make no claim that AI literacy causes reduced adoption of any tool. The mediation analysis in the original paper (perceptions of AI as magical) is not addressed here and remains the authors’ primary theoretical mechanism.
- No test of magicalness by tool type. An obvious extension would be to ask whether perceived “magicalness” varies systematically across non-text and text tool categories, and whether that variation can account for the asymmetry we document. This would require new data.
- Clustering. Our item-level standard errors do not cluster by participant. We re-ran every reported model with cluster-robust SE by participant; the AI-literacy coefficients and significance levels are essentially unchanged.

8 Conclusion

We do not dispute the aggregate lower-literacy/higher-usage association reported by [Tully et al. \(2025\)](#); rather, we show that its substantive meaning changes when AI usage is decomposed by tool type and modelled as ordinal adoption data. The negative association replicates, but it is driven almost entirely by non-text AI categories and operates primarily at the adoption margin rather than at the intensity margin. We therefore read Study 3 as evidence about non-text AI adoption breadth rather than as evidence about general AI receptivity or text-AI use. This qualification preserves the spirit of the original contribution while sharpening its managerial and theoretical implications.

Data and code availability

The Study 3 data are publicly available from the original authors at [ResearchBox #1491](#). All Python code used to produce the tables and figures in this paper, together with a fully executed Jupyter notebook, is available in the project repository: [HristoInouzhe/AI-use-vs-AI-literacy](#).

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